Computer Vision
Lecture 13: Morphological Operation

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Mathematical morphology

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Jean Serra







Mathematical morpho

- Mines ParisTech
- 1783년 3월 19일 루이 16세 정부의 자문회의의 법령에 의해서 왕실의 광산을 이끌어갈 공학자들을 훈련시키기 위해 설립되었다



Set Theory

영남대학교

A,B = Set of pixels = set of coordinates

$A \cup B = \{ x \mid x \in A \text{ or } x \in B\}$ Union

$A \cap B = \{ x \mid x \in A \text{ and } x \in B\}$ Intersection

$A^c = \{ x \mid x \in A\}$ Complement

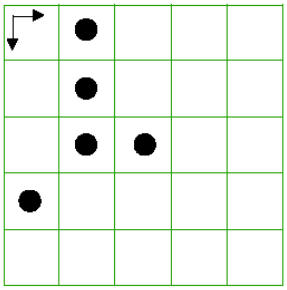
$A - B = \{ x \mid x \in A , x \in B\}$ Difference

$(A)_c = \{ x \mid x=a+c , a \in A\}$ Translation

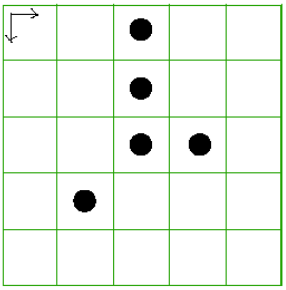
$A^{\wedge} = \{ x \mid -x \in A\}$ Inversion

Translation

$$(A)_x = \{c | c = a + x, \ for \ a \in A\}$$



A



$(A)_x$

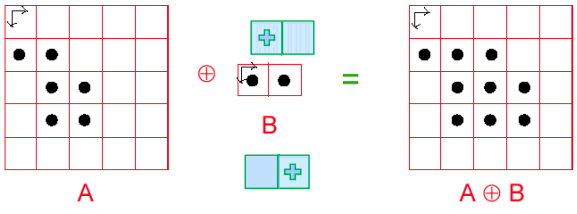
A = {(1,0), (1,1), (1,2), (2,2), (0,3)}

X = {(1,0)}

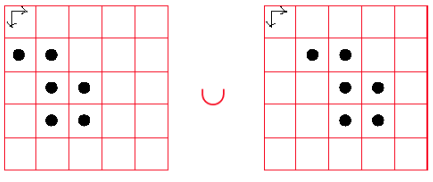
$(A)_X$ = {(2,0), (2,1), (2,2), (3,2), (1,3)}

Dilation

$$A \oplus B = \bigcup_{b \in B} (A)_b = \{x | (\hat{B})_x \cap A \neq \phi\}$$



- B: Structuring element



A = {(0,1), (1,1), (1,2), (2,2), (1,3),(2,3)}

B = {(0,0), (1,0)} : Structuring element

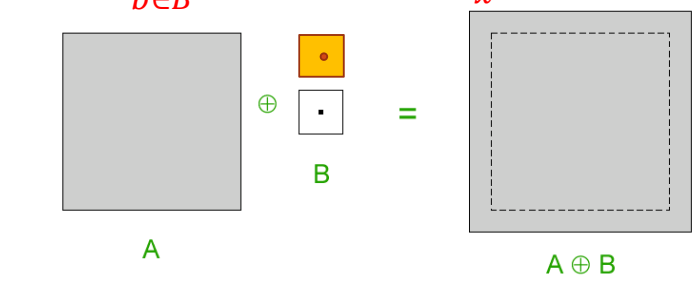
$(A)_{(0,0)}$ = {(0,1), (1,1), (1,2), (2,2), (1,3),(2,3)}

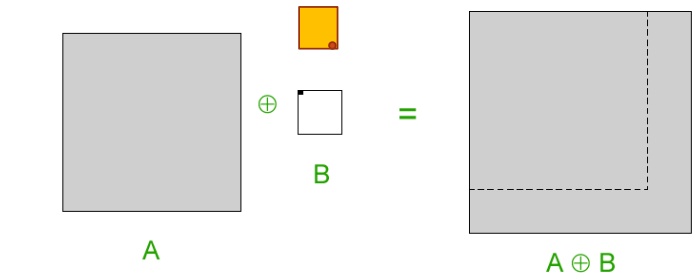
$(A)_{(1,0)}$ = {(1,1), (2,1), (2,2), (3,2), (2,3),(3,3)}

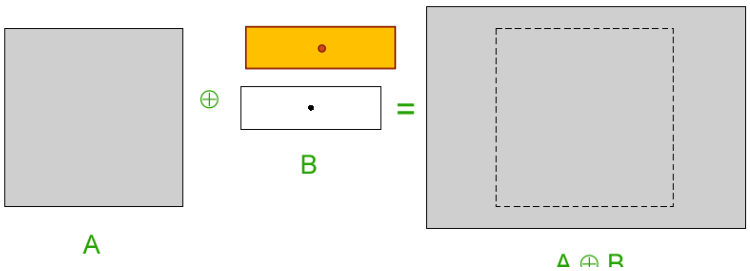
$U(A)_B$ = {(0,1), (1,1), (1,2),(2,1),(2,2), (3,2),(1,3),(2,3),(3,3)}

Dilation

$$A \oplus B = \bigcup_{b \in B} (A)_b = \{x | (\hat{B})_x \cap A \neq \phi\}$$



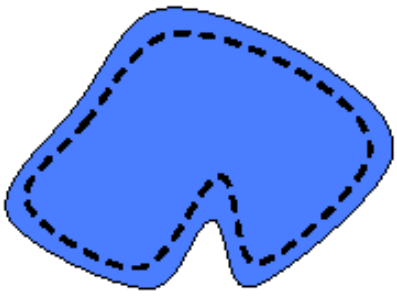




Dilation

$$A \oplus B = \bigcup_{b \in B} (A)_b = \{x | (\hat{B})_x \cap A \neq \phi\}$$





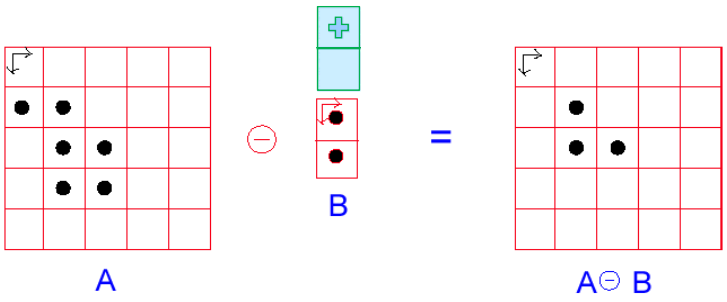
Dilation Properties

$$A \oplus B = \bigcup_{b \in B} (A)_b = \{x | (\hat{B})_x \cap A \neq \phi\}$$

$A \oplus B = B \oplus A$	Commutative
$A \oplus (B \oplus C) = (A \oplus B) \oplus C$	Associative
$A \subseteq B \Rightarrow C \oplus A \subseteq C \oplus B$	Monotone
$(A \cup B) \oplus C = (A \oplus C) \cup (B \oplus C)$	Distributive
$(A \cap B) \oplus C \subseteq (A \oplus C) \cap (B \oplus C)$	

Erosion

$$A \ominus B = \bigcap_{b \in B} (A)_{-b} = \{x | (B)_x \subseteq A\}$$



A = {(0,1), (1,1), (1,2), (2,2), (1,3),(2,3)}

B = {(0,0), (0,1)} : Structuring element

B^ = {(0,0), (0,-1)}

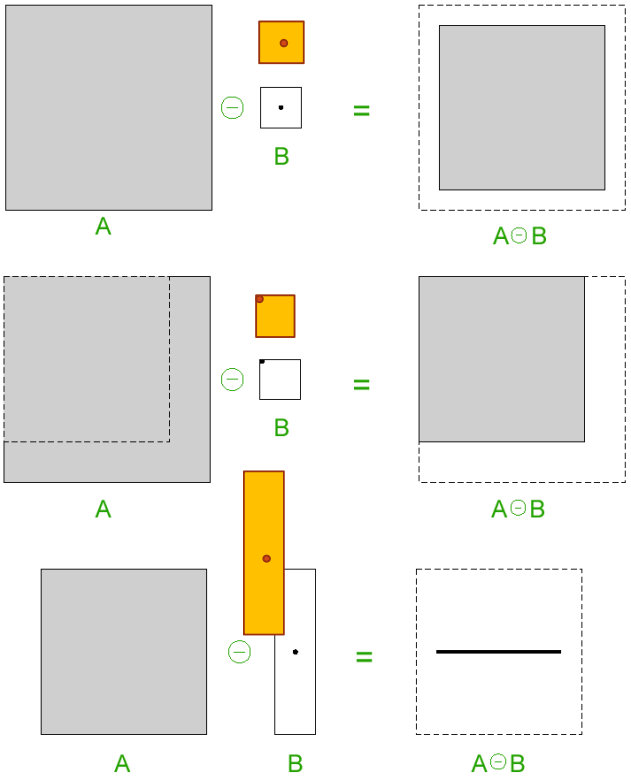
$(A)_{(0,0)}$ = {(0,1), (1,1), (1,2), (2,2), (1,3),(2,3)}

$(A)_{(0,-1)}$ = {(0,0), (1,0), (1,1), (2,1), (1,2),(2,2)}

$\cap (A)_{B^\wedge}$ = {(1,1),(1,2),(2,2)}

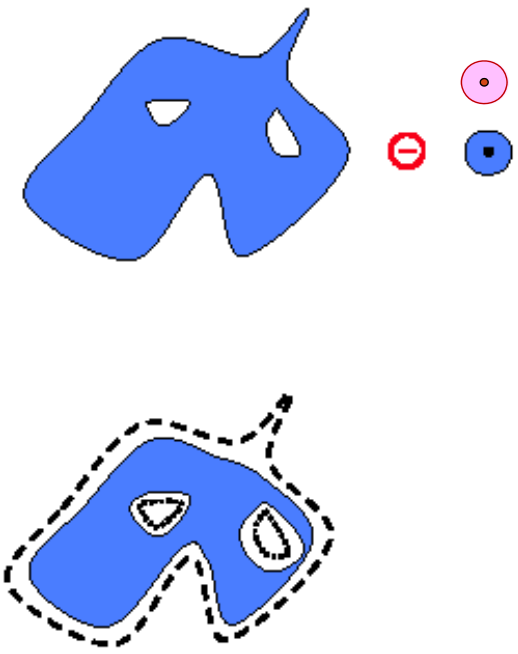
Erosion

$$A \ominus B = \bigcap_{b \in B} (A)_{-b} = \{x | (B)_x \subseteq A\}$$



Erosion

$$A \ominus B = \bigcap_{b \in B} (A)_{-b} = \{x | (B)_x \subseteq A\}$$



Erosion application Distance Transform

$$A \ominus B = \bigcap_{b \in B} (A)_{-b} = \{x | (B)_x \subseteq A\}$$

- Simple but inefficient
- Multiple successive erosions with suitable stucturing element until all foreground regions have been eroded away
- If each pixel is labeled with the number of erosions that had to be performed before it disappeared, then this is just the distance transform.

0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

⇒

0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	2	2	2	2	1	0
0	1	2	3	3	2	1	0
0	1	2	2	2	2	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

Erosion Properties

$$A \ominus B = \bigcap_{b \in B} (A)_{-b} = \{x | (B)_x \subseteq A\}$$

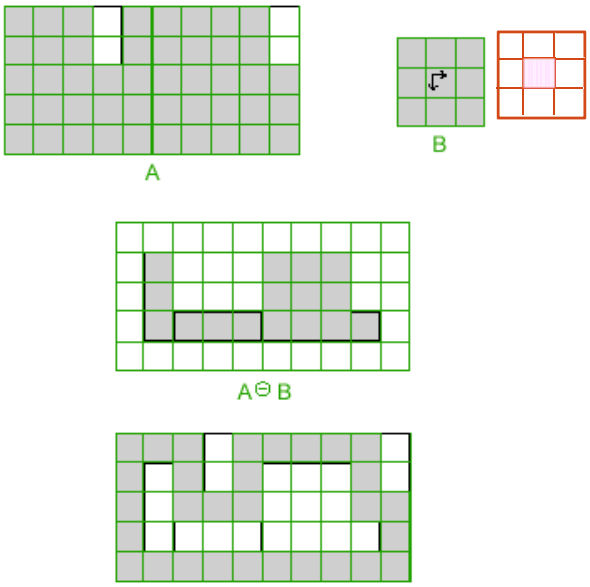
$A \subseteq B \Rightarrow A \ominus C \subseteq B \ominus C$ Monotone

$A \supseteq B \Rightarrow C \ominus A \subseteq C \ominus B$

$(A \ominus B)^c = A^c \oplus \hat{B}$ Erosion vs Dilation

Application: Edge Detection using Erosion

$$\text{Edge}(A) = A - (A \ominus B)$$



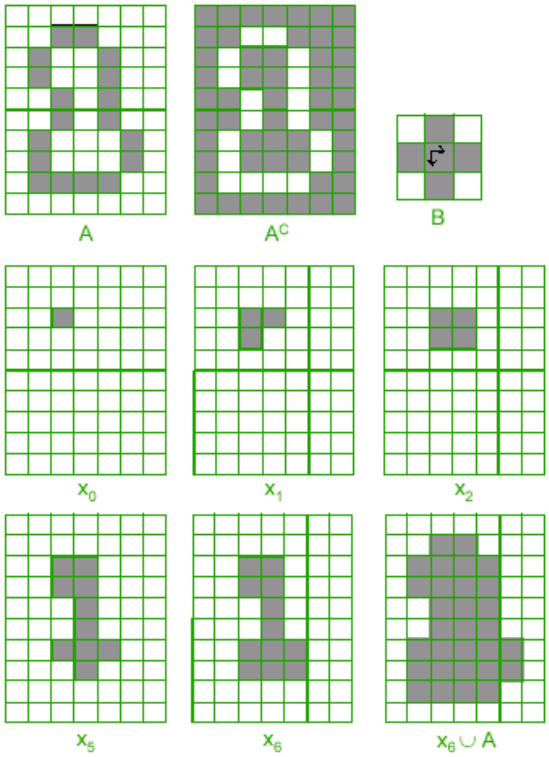
Application: Edge Detection using Erosion



$$\text{Edge}(A) = A - (A \ominus B)$$

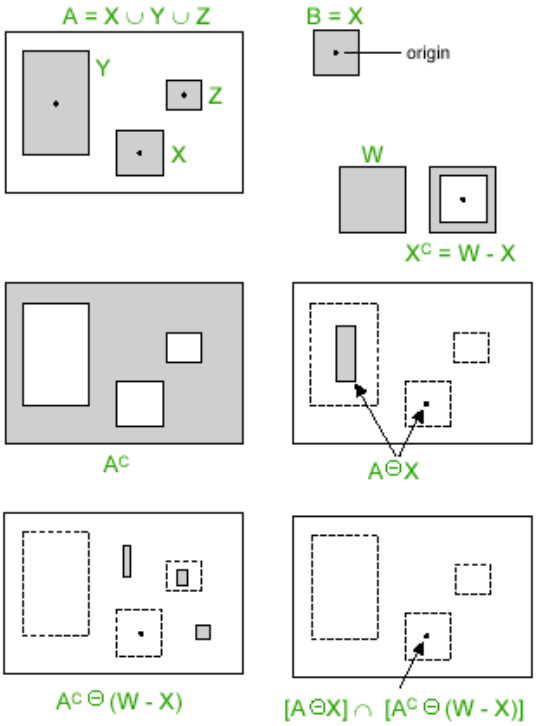
Application: Region Filling

$$X_k = (X_{k-1} \oplus B) \cap A^c$$



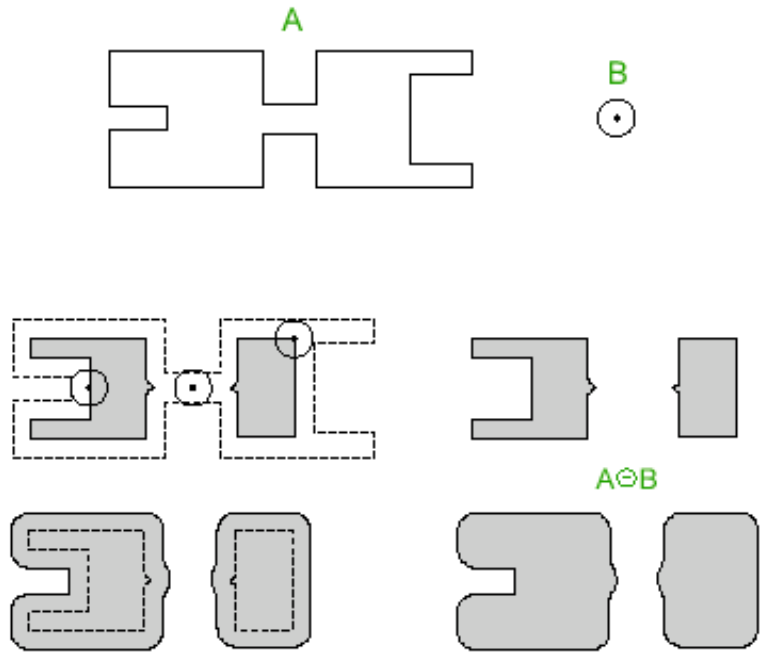
Application: Hit or Miss

$$A \circledast B = (A \ominus B) \cap (A^c \ominus B^c)$$



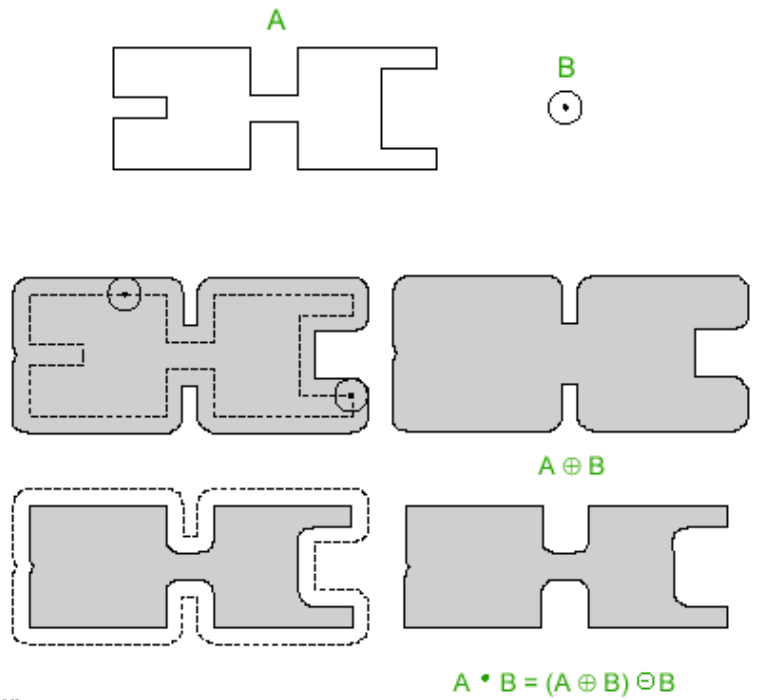
Opening

$$A \circ B = (A \ominus B) \oplus B$$



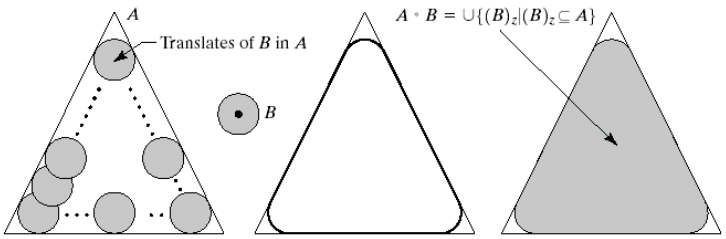
Closing

$$A \bullet B = (A \oplus B) \ominus B$$



Opening vs. Closing

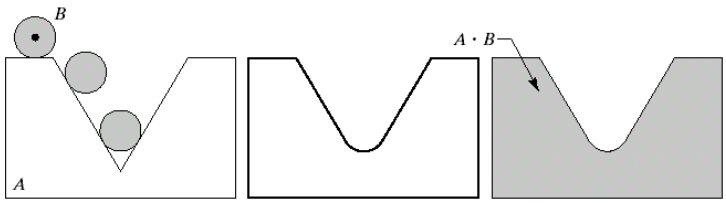
Opening



a b c d

FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded).

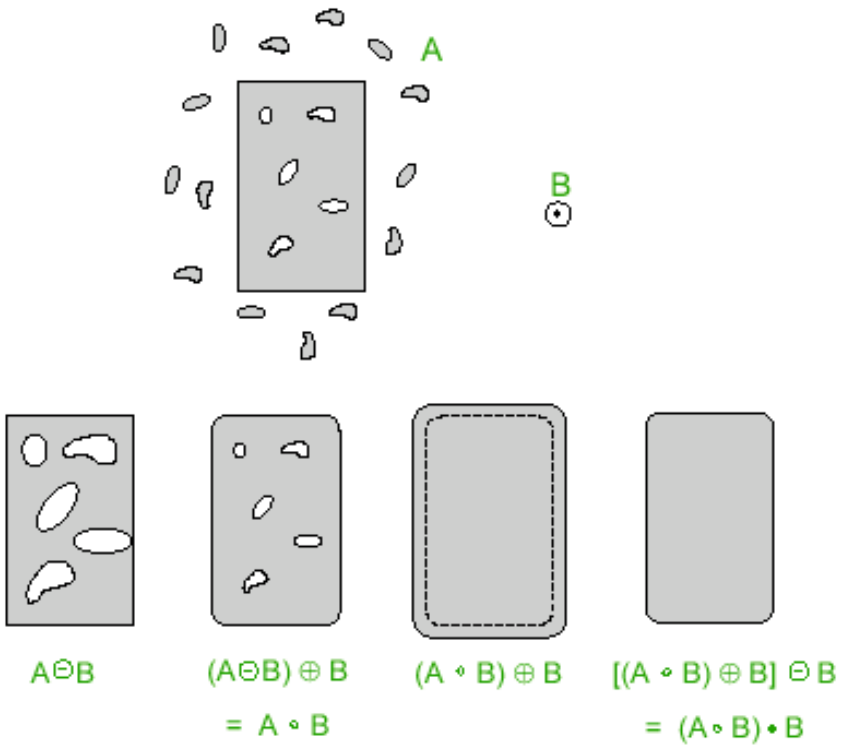
Closing



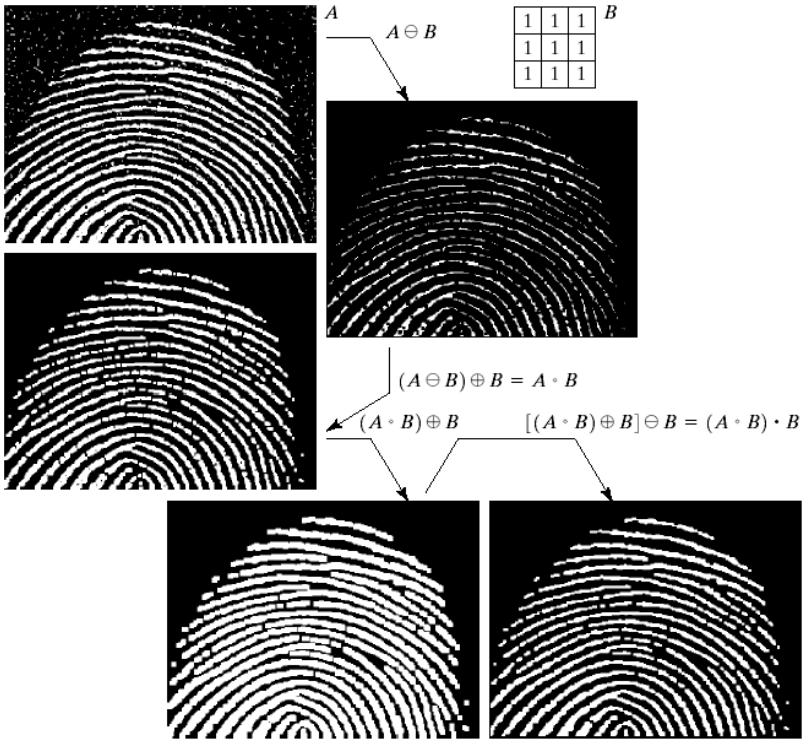
a b c

FIGURE 9.9 (a) Structuring element B "rolling" on the outer boundary of set A. (b) Heavy line is the outer boundary of the closing. (c) Complete closing (shaded).

Application: open-closing



Application: open-closing



a b
d c
e f

FIGURE 9.11
(a) Noisy image. (c) Eroded image. (d) Opening of A. (d) Dilation of the opening. (e) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)

Grayscale Morphological Operation

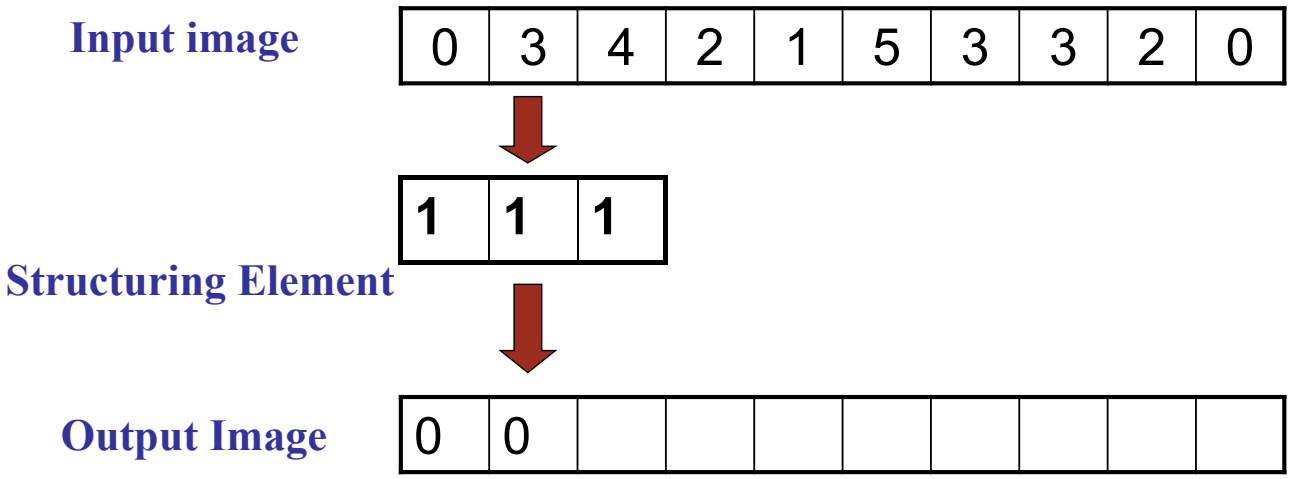
Dilation

$$(f \oplus b)(s) = \max\{f(s-x) + b(x) | (s-x) \in D_f and x \in D_b\}$$

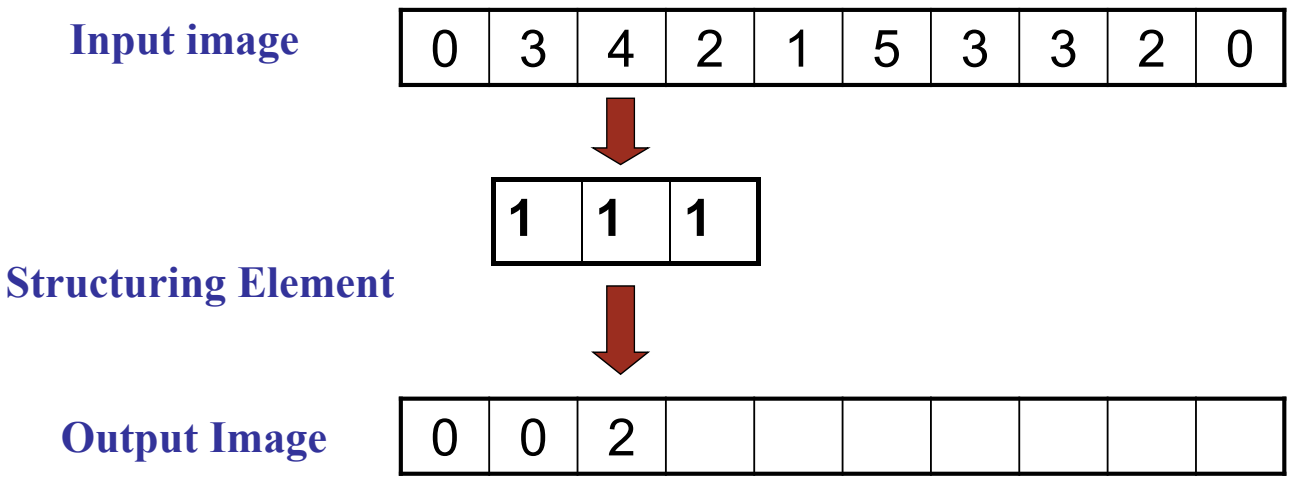
Erosion

$$(f \ominus b)(s) = \min\{f(s+x) - b(x) | (s+x) \in D_f and x \in D_b\}$$

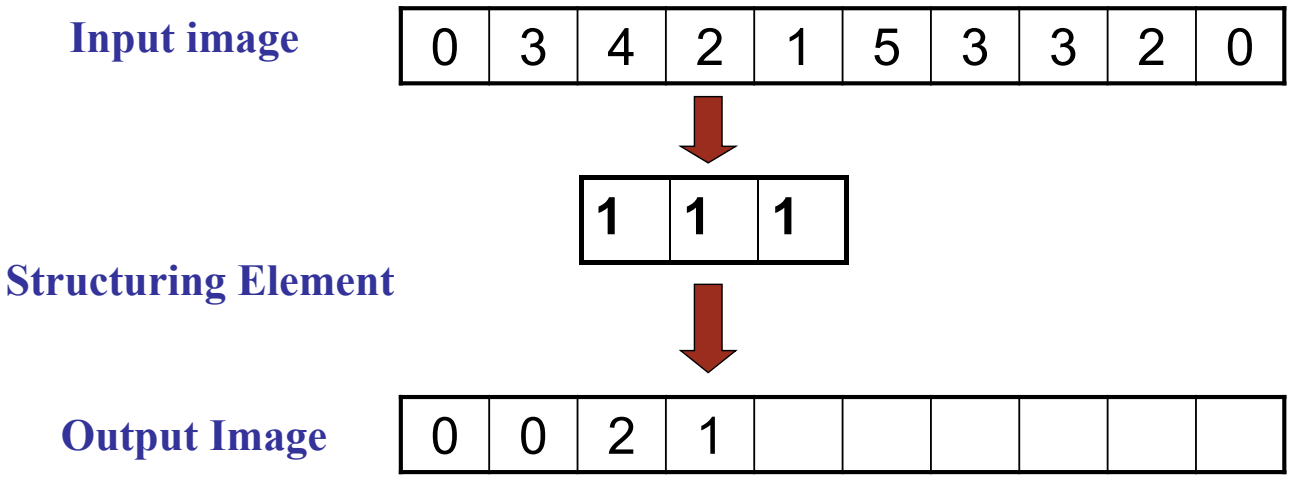
Example of 1D Grayscale Erosion Operation



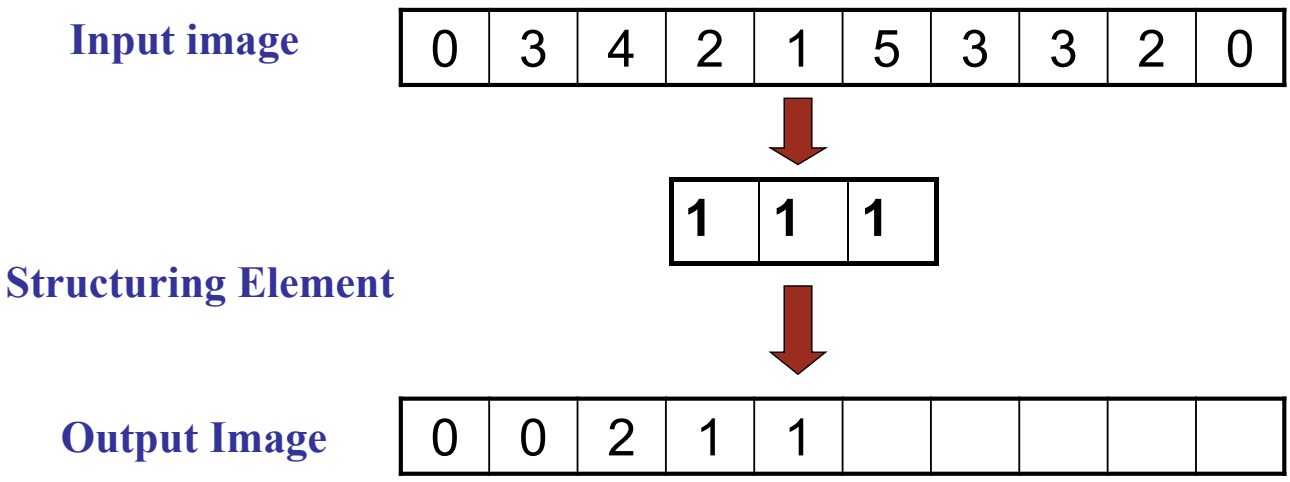
Example of 1D Grayscale Erosion Operation



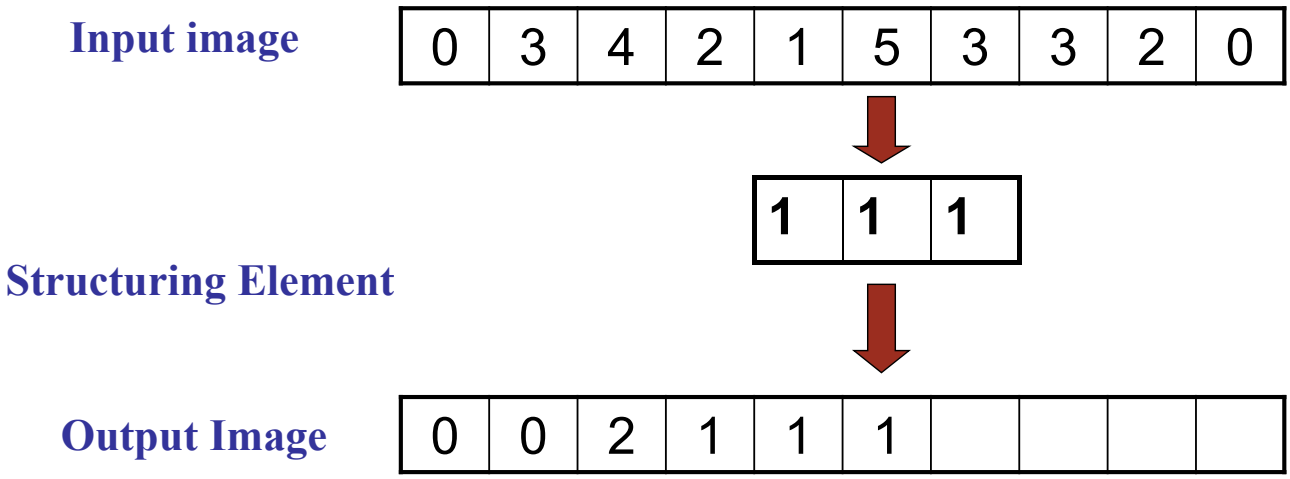
Example of 1D Grayscale Erosion Operation



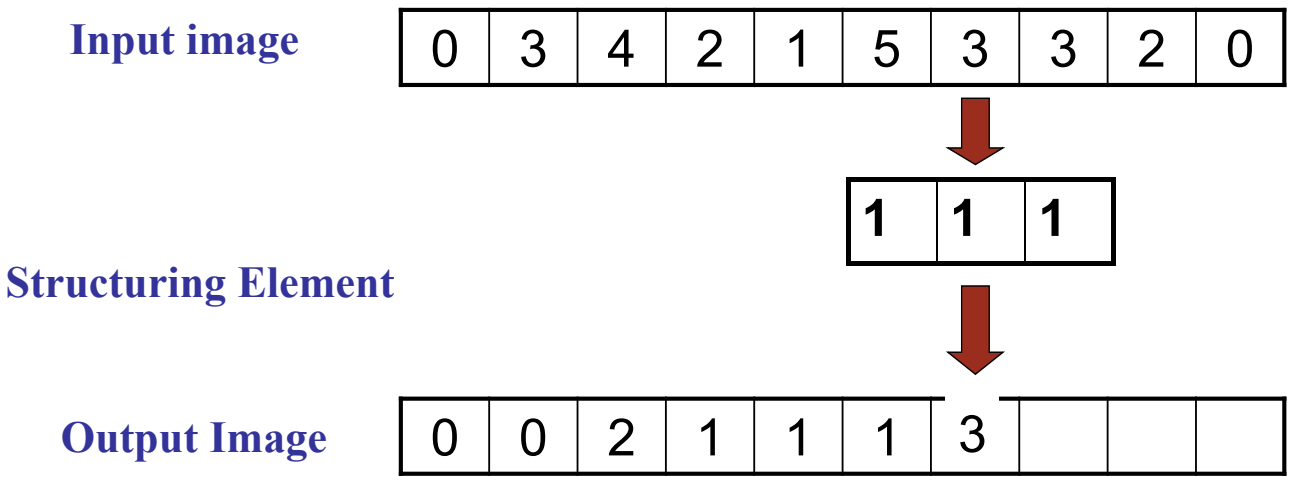
Example of 1D Grayscale Erosion Operation



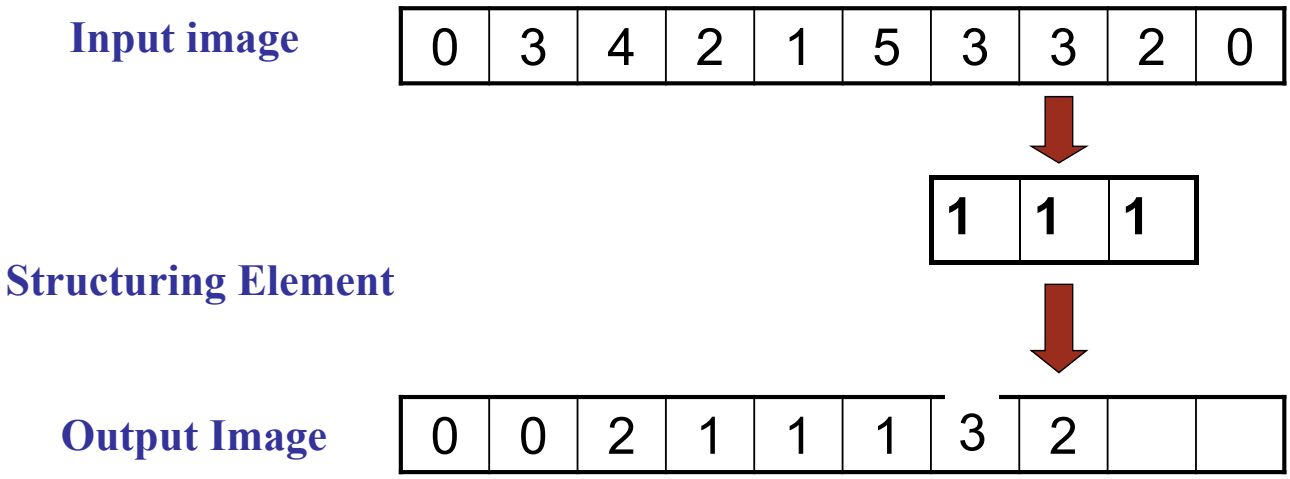
Example of 1D Grayscale Erosion Operation



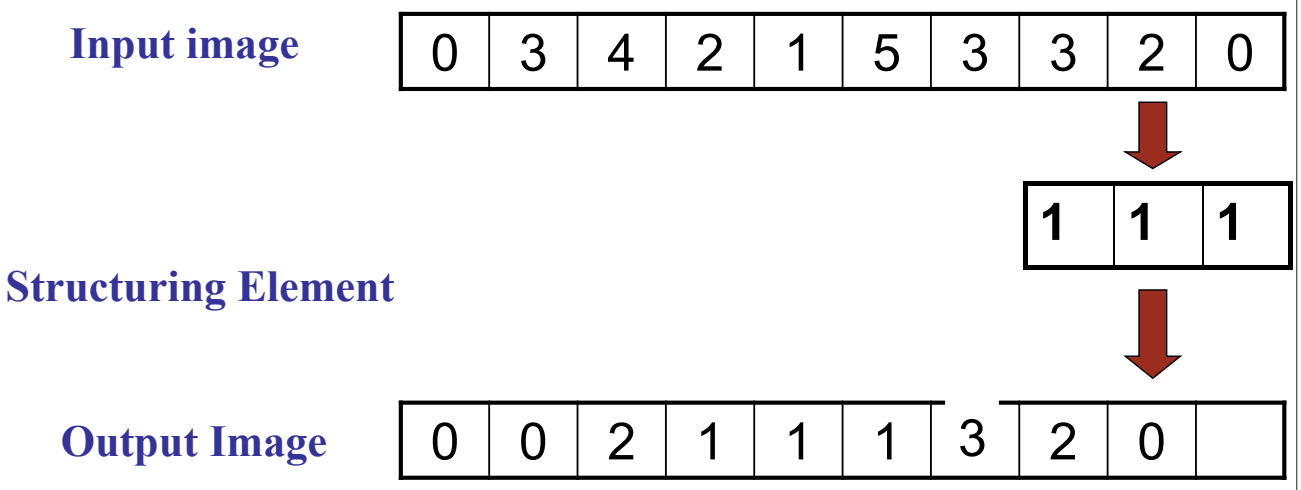
Example of 1D Grayscale Erosion Operation



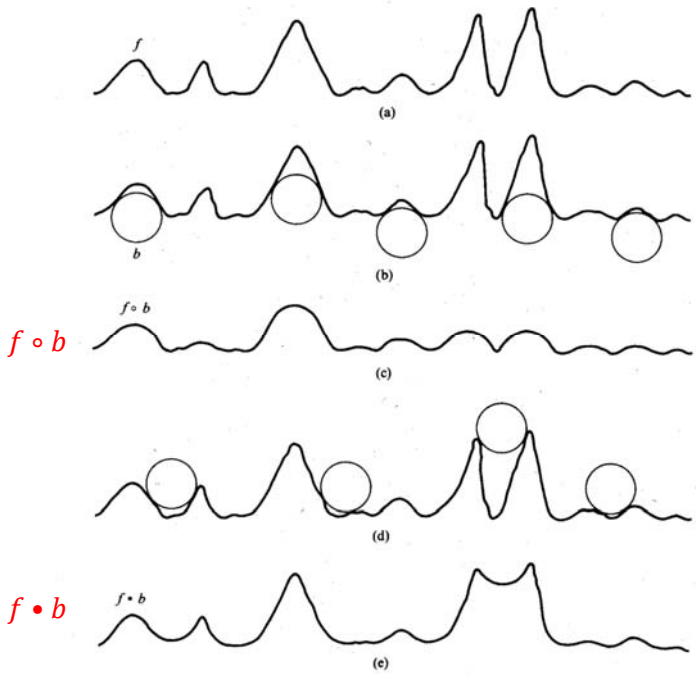
Example of 1D Grayscale Erosion Operation



Example of 1D Grayscale Erosion Operation



Grayscale Morphological Operation



Grayscale Morphological Operation



Dilation



Erosion



Opening



Closing

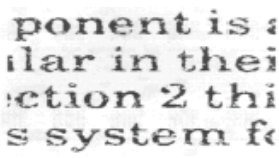
Applications: Noise Filtering

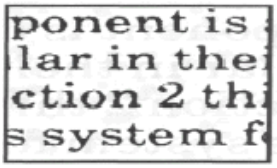


Original + Noise

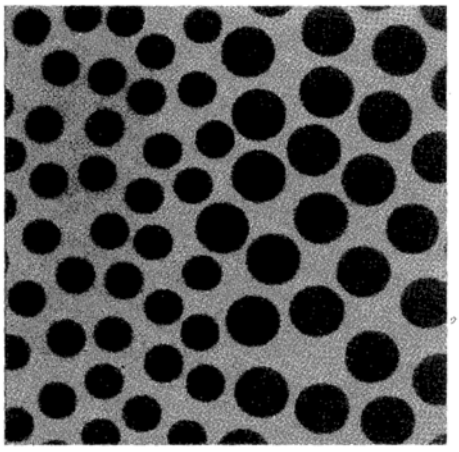


Opening + Closing

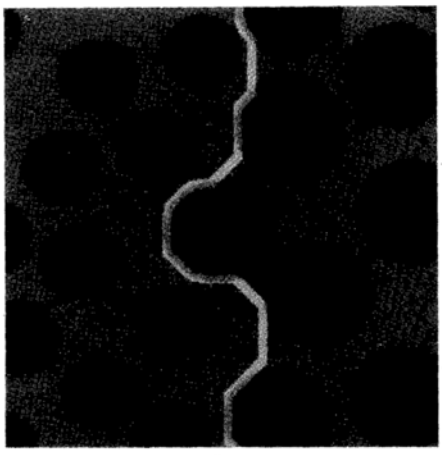




Applications: Segmentation

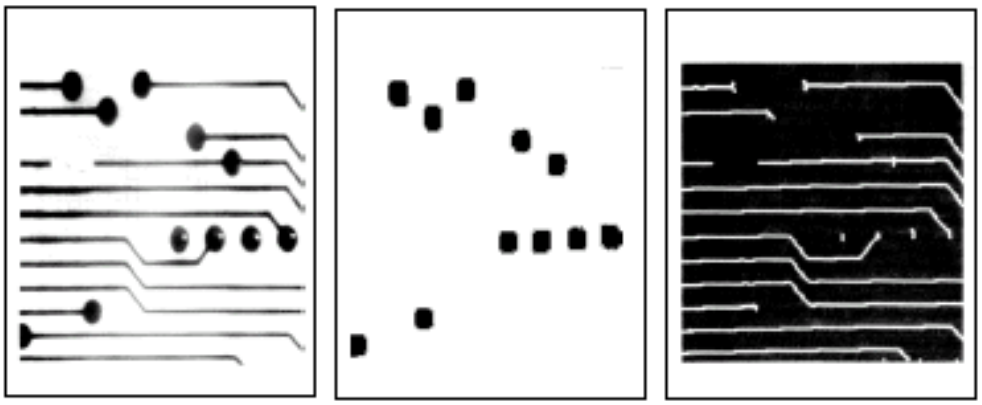


(a)



(b)

Applications: PCB



Printed Circuit Board | Opening | Skeleton Pruning

Thank You
Next : Convolution

